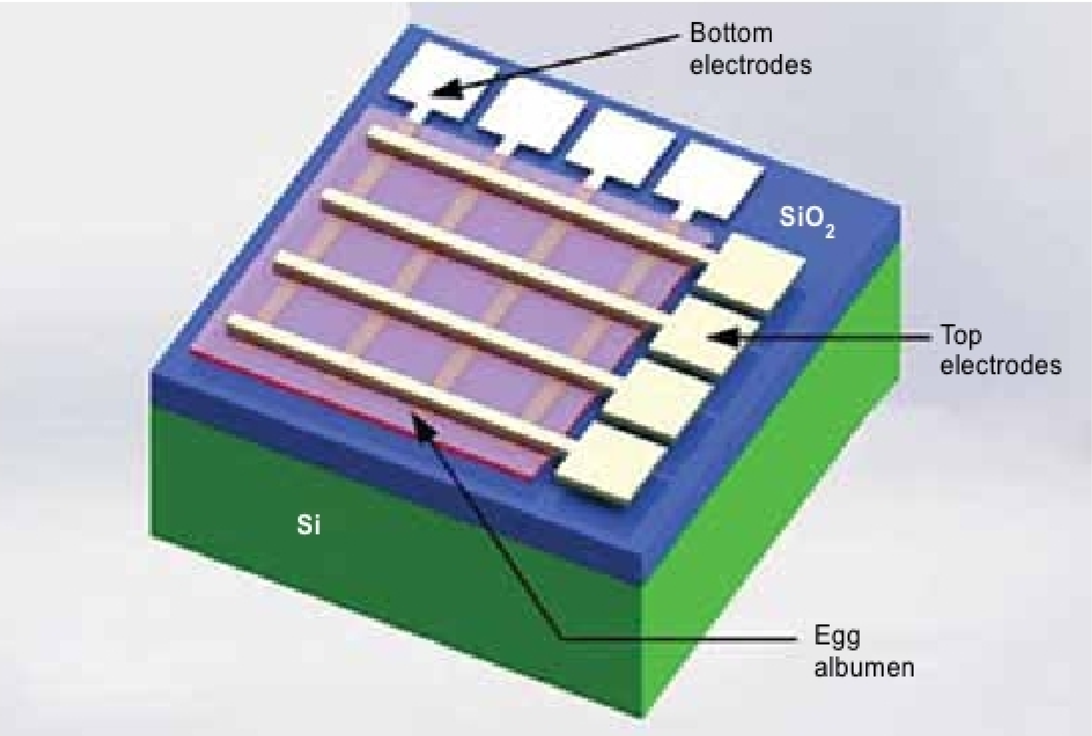
Egg-Based Electronics To Save The Environment

Organic electronic devices have recently attracted great attention because these offer several advantages such as simple device structure, potentially low cost, and low power consumption. Researchers are currently working on egg white to build organic electronics as it is easy to process, readily accessible, and available at a lower cost



Egg-based memristor (Credit: www.eenewseurope.com)

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Durability is often seen as a hallmark for high-quality electronic systems, but sometimes we need components that don't last for a long period. For example, it would be handy if microelectronic systems that delivered drugs to various parts of the body dissolved after their tasks were over. Similarly, there are many electronic devices like sensors that should simply get dissolved after finishing their job rather than contaminating the environment. The use of electronic devices has grown exponentially, and the United Nations estimates that people throw away about fifty million metric tons of electronics every year.

One way to lessen the problem, some scientists say, is to use biological materials to build devices that are biodegradable and biocompatible. Using biological materials to build electronics would be environment friendly because their ability to biodegrade would lead to less piling up of trash. Moreover, biodegradable materials are non-toxic.

These materials have earned considerable attention in recent years. Biomaterials have the potential to cut the cost of organic electronics and simplify manufacturing processes.

Biomaterials are already used as semiconductors, dielectrics, and substrates in bio-organic field-effect transistors (BiOFETs), but these materials are not always easy to handle and may require many extraction steps. Egg, a well-known dietary and cosmetic supplement, is now advocated as biodegradable, biocompatible, and environment-friendly material towards bioelectronics.

Therefore egg albumen can be a promising alternative to the conventional silicon-based non-volatile memory systems. It may also have potential applications in the imitation of biological memory behaviour, artificial intelligence, and brain-like intelligence because of the good compatibility.

Researchers have recently figured out how to create one such chip out of eggs. They demonstrated that egg albumen-based resistive memory compares favourably to other memories, exhibiting a high on/off resistance ratio as well as good retention and switching endurance even after repeated bending.

Organic electronics

Organic-based optoelectronic and electronic devices have recently attracted great attention because of their potential applications in lighting systems, flat panel displays, backlighting modules, radio frequency identification tags, flexible sensors, and flexible displays.

These organic-based devices offer numerous advantages such as simple device